

DTSA 5750

Numerical Methods for Sampling Probability Distributions

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Data Scientist (37.3 / 100)	28.7 / 100
SPECIALIZATION Simulation Methods for Data Science		

29

CURRENT AVG SCORE
across 6 job roles

63

POTENTIAL SCORE
with recommended additions

+42

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will cover the mathematics and numerical details of using Hamiltonian dynamics and Markov Chain Monte Carlo methods to generate samples from probability distributions. It will cover both sampling of physical systems as well as statistical problems such as sampling the posterior and parameter estimation in Bayesian analysis. Both theory and practical strategies for performing efficient calculations and analyzing their results will be discussed. Exercises and examples will be in Python.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Data Scientist	37.3	78.8	+41.5	HIGH
Applied Scientist	34.5	68.5	+34.0	HIGH
Machine Learning Engineer	30.5	70.4	+39.9	HIGH
AI Engineer	25.7	57.9	+32.1	HIGH
ML Engineer	23.2	51.9	+28.6	MEDIUM
MLOps Engineer	21.0	53.3	+32.3	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning

1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

Data Scientist	37.3 → 78.8 (+41.5 pts)
data sets behaviors social build approaches	
Applied Scientist	34.5 → 68.5 (+34.0 pts)
biology computational models data learning design real fundamental	
Machine Learning Engineer	30.5 → 70.4 (+39.9 pts)
data sets learning tools exploratory design science models	
AI Engineer	25.7 → 57.9 (+32.1 pts)
data models learning build software model machine design	
ML Engineer	23.2 → 51.9 (+28.6 pts)
data learning machine models model build global tools	
MLOps Engineer	21.0 → 53.3 (+32.3 pts)
data model learning machine training tools models error	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.